IAEA nuclear fusion

Voltage characteristics of a rotating plasma

To cite this article: B. Lehnert 1961 *Nucl. Fusion* **1** 125

NUCLEAR FUSION 1 (1961) 125-130

VOLTAGE CHARACTERISTICS OF A ROTATING PLASMA

B. Lehnert

Royal Institute of Technology, Stockholm, Sweden

The balance of forces is investigated for a rotating plasma contained in a long cylindrical vessel and placed in an axial magnetic field. A relation is deduced between the voltage which arises in a radial direction in the plasma and the electric charge being transmitted through the system from an external source. It is found that the equivalent capacity, which describes the electrical behaviour of the plasma, depends upon the transmitted charge and upon the constitution of the vessel wall. An explanation is suggested for the voltage limiting effect earlier observed in the Ixion device.

1. Introduction

The study of a rotating magnetized plasma is of importance both in astrophysics and in research on controlled thermonuclear fusion. Most of the configurations discussed so far are axially symmetric and include a poloidal magnetic field in planes through the axis of symmetry. The field confines the plasma in a radial direction and guides it in the rotation around the axis. The present paper concerns the situation where the rotation is produced by an electric current which is generated by external sources and which flows across the magnetic field. Arrangements of this kind have earlier been reported by a number of authors [1]—[4] (see also [5]).

It has been found by Alfvén [6] that a plasma moving across a magnetic field behaves in the same way as an equivalent capacity where the stored energy corresponds to the kinetic energy of the motion. Both Anderson *et al.* [2, 3] and Boyer *et al.* [4] point out that this capacity will be modified in a rotating plasma when the centrifugal force becomes large enough to deform the magnetic field. Part of the energy fed into the system will then be used to increase the potential energy of the field.

In the present paper the distortion of the magnetic field and its influence on the equivalent capacity will be investigated. In particular, the situation will be studied where the energy is fed into the system from a condenser bank. This gives information about the voltage characteristics which describe the relation between the voltage of the bank before connection to the plasma device and the voltage across the plasma when full rotation has been established. Due to the deformation of the magnetic field the equivalent capacity will depend upon the voltage applied across the device and upon the constitution of the walls of the discharge chamber. As a consequence, the characteristics will not become linear and the voltage across the plasma may sometimes have a tendency to saturate when the voltage of the energy source is increased.

These phenomena will be studied in the following sections for the simplified situation of a long, cylindrical configuration.

2. Theory on the balance of forces

2.1 Basic equations and approximations

Consider a quasi-neutral, dissipation-free plasma of density ϱ where the centre of mass of a fluid element moves with the velocity $\mathbf{v}$ and the electric current density is $\mathbf{i}$. Conservation of mass and charge is expressed by

$$\operatorname{div} \mathbf{v} = -\frac{1}{\varrho} \cdot \frac{d\varrho}{dt} \tag{1}$$

and

$$\operatorname{div} \mathbf{i} = 0, \tag{2}$$

where $d/dt = \partial/\partial t + \mathbf{v} \cdot \nabla$. Conservation of momentum is represented by the equation of motion

$$\varrho \frac{d\mathbf{v}}{dt} = \mathbf{i} \times \mathbf{B} - \nabla p \tag{3}$$

and by Ohm's law

$$0 = \mathbf{E} + \mathbf{v} \times \mathbf{B} - \frac{m_i - m_e}{e\varrho} \mathbf{i} \times \mathbf{B} + \frac{1}{e\varrho}(m_i \nabla p_e - m_e \nabla p_i), \tag{4}$$

where $\mathbf{B}$ is the magnetic field, $\mathbf{E}$ the electric field and p is the sum of the ion pressure p_i and the electron pressure p_e. The rest of the symbols used in Eq. (4) have their conventional meaning. In Eqs. (3) and (4) the time changes are assumed to be slow enough and the radii of gyration small enough for terms containing $\mathbf{i} \cdot \nabla$ and $d(\mathbf{i}/\varrho)/dt$ to be neglected.

To the system of Eqs. (1)—(4) are added Maxwell's equations

$$\operatorname{curl} \mathbf{B} = \mu_0 \mathbf{i} \tag{5}$$

$$\operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{6}$$

where μ_0 indicates the magnetic permeability in vacuo and MKS-units are used.

Introduce the velocity

$$\mathbf{v}^* = \mathbf{v} - (m_i - m_e)\, \mathbf{i}/e\varrho \tag{7}$$

and substitute $\mathbf{E}$ from Eq. (4) into Eq. (6). Observing that div $\mathbf{B} = 0$ and using Eqs. (1) and (2) together with well-known vector operations for curl $(\mathbf{v}^* \times \mathbf{B})$, the result becomes

$$\left(\frac{\partial}{\partial t}+\mathbf{v}^*\cdot\nabla\right)(\mathbf{B}/\varrho) = \left(\frac{\mathbf{B}}{\varrho}\cdot\nabla\right)\mathbf{v}^* + \frac{1}{e\varrho^3}(m_i\nabla p_e - m_e\nabla p_i)\times\nabla\varrho. \quad (8)$$

When $\mathbf{v}^*=\mathbf{v}$ and the pressure terms vanish, Eq. (8) becomes identical with a relation earlier deduced by WALÉN [7]. Observe that the pressure terms vanish when the pressure and density gradients are parallel.

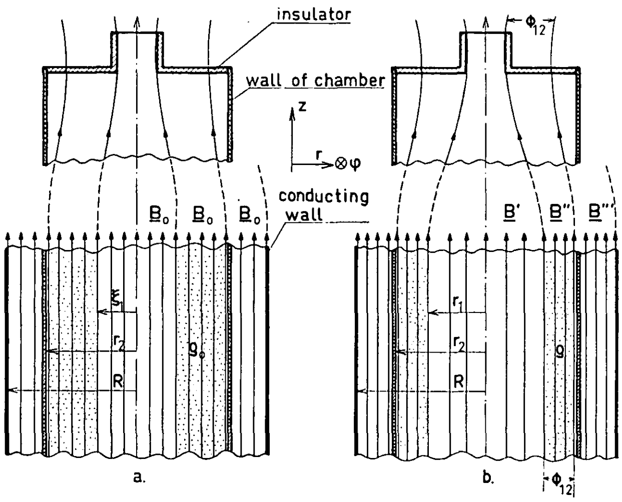

Fig. 1 Axially symmetric configuration with axial magnetic field and radial electric field which set the plasma into rotation in the φ-direction within the dotted regions of the figure. The upper parts of the figure show the ends of the configuration and the lower parts the central region which is assumed to be situated far from the ends. a—situation at the beginning of the rotation; b—situation when a radial expansion is caused by the centrifugal force. The voltage Φ_{12} across the rotating plasma is applied between the cylindrical wall of the chamber and the edge of the hole of the insulator.

The discussion will now be restricted to a simple cylindrically symmetric configuration where the following assumptions are valid and apply to Fig. 1:

(i) All field quantities are independent of φ and z in a cylindrical co-ordinate system (r, φ, z) with z along the axis of symmetry.

(ii) Initially there exists a homogeneous magnetic field $\mathbf{B}_0$ along the z-axis within the space to be studied by the present theory. The region occupied by the plasma has a constant initial density ϱ_0.

(iii) An electric field component E_r is applied in the radial direction which creates a total current density i_r and sets the plasma into rotation around the axis. This gives rise to a centrifugal force, a radial expansion, an induced electric field E_φ and a current i_φ.

(iv) The radial expansion velocity v_r of the centre of mass of the plasma is much smaller than the velocity v_φ of rotation.

(v) The magnetic field generated by the current i_r is neglected; only the field arising from i_φ is taken into account. Consequently, the resulting magnetic field $\mathbf{B}$ will be directed along the z-axis.

With these assumptions the basic equations (3), (4), (5) and (8) reduce to

$$\varrho v_\varphi^2/r = (B/\mu_0)\frac{\partial B}{\partial r} + \frac{\partial p}{\partial r}, \quad (9)$$

$$\varrho\frac{\partial v_\varphi}{\partial t} + \varrho v_r\left(\frac{\partial v_\varphi}{\partial r}+\frac{v_\varphi}{r}\right) = -i_r B, \quad (10)$$

$$0 = E_r + v_\varphi B - \frac{m_i - m_e}{e\varrho} i_\varphi B + \frac{1}{e\varrho}\left(m_i\frac{\partial p_e}{\partial r} - m_e\frac{\partial p_i}{\partial r}\right), \quad (11)$$

$$B/\varrho = B_0/\varrho_0. \quad (12)$$

For a rough estimate of orders of magnitude, space derivatives can be approximated by inverted characteristic lengths. In doing so, it is found that the ratio between the pressure term in Eq. (9) and the left hand member of the same equation is of the order of the square of the ratio between the thermal velocity of the ions and the velocity of rotation v_φ. The result is valid when the characteristic length of the pressure variation is of the order of the dimensions of the configuration and when the electron temperature does not exceed that of the ions by very much. The following discussion will be simplified by assuming a temperature which is low enough for the pressure gradient to be neglected in Eq. (9).

Further, the magnitude of the third and fourth terms of the right hand member of Eq. (11) is likely to become small compared to that of the second term of the same member. This occurs when the gyro-period of an ion is much shorter than the time required for the mass flow to make one revolution around the axis and when the thermal energy of the particles is much less than the energy corresponding to the radial potential difference across the whole configuration.

2.2 Voltage characteristics and equivalent capacity.

When the terms containing the radial velocity v_r in Eq. (10) are neglected, combination with Eq. (12) gives

$$v_\varphi = -B_0 Q(t)/2\pi r\varrho_0, \quad (13)$$

where the electric charge which has been transmitted in a radial direction per unit axial length is

$$Q(t) = \int_0^t 2\pi r i_r\,dt \quad (14)$$

and the initial condition $v_\varphi(0)=0$ is imposed. It should be pointed out that the terms containing v_r in Eq. (10) cancel when v_φ has the form (13). Consequently, Eq. (13) is a good approximation not only for cases where the radial displacements are small compared with the distance r from the axis. Also observe that the velocity of rotation in Eq. (13) is independent of the state of deformation of the magnetic field as well as of the pressure gradients in Eqs. (9) and (11). Since viscous dissipation has been neglected, the result that v_φ is inversely proportional to r does not contradict the equations of motion. The omission of viscous effects provides a good approximation at high temperatures and in strong magnetic fields where viscous interaction between the fluid layers is strongly reduced in the radial direction. It should also be observed that the force $\eta \nabla^2 v + (1/3) \eta \nabla (\mathrm{div}\, \mathbf{v})$ due to dynamic viscosity η vanishes for a velocity field $\mathbf{v} = 0$, const./r, 0.

When the pressure term in Eq. (9) is dropped, according to the assumptions of the preceding paragraph, the magnetic field in the plasma obeys the equation

$$\frac{\partial B}{\partial r} = 2\, B_0 \frac{a^2}{r^3}\,; \qquad a^2 = \frac{\mu_0 Q^2}{8 \pi^2 \varrho_0}. \tag{15}$$

This is deduced from Eqs. (13), (9) and (12). Thus,

$$B = B_0\, (b'' - a^2/r^2) \tag{16}$$

in the plasma region where b'' is a constant of integration. The magnetic flux contained between surfaces at r_m and r_n is

$$\Phi_{mn} = \int_{r_m}^{r_n} 2\pi r\, B\, dr = \pi B_0\, b''\, (r_n^2 - r_m^2) - 2\pi\, B_0 a^2 \log (r_n/r_m)\,. \tag{17}$$

The problem is now specified by considering the configuration of Fig. 1 which is closely related to the conditions of the Ixion device [4]. Assume a very long cylindrical vacuum chamber where the wall of radius r_2 is used as one electrode. The ends of the chamber are arranged as shown by the upper half of Figs. 1a. and 1b. and are assumed to be far away from the central region shown by the lower half of the same figures. The ends are provided by a pair of hat-shaped insulators. Plasma is injected and fills the space of the chamber. A voltage is applied between the wall at r_2 and the central part of the plasma which is projected through the hole in the insulators as shown. The part of the plasma shown by the dots in Figs. 1a. and 1b. will then be set into rotation and the voltage across the rotating part will be given by Φ_{12}. The potential applied at the edge of the insulator hole will project itself along a magnetic field line to a surface of radius ξ_1 in the lower part of Fig. 1a. This figure shows the situation before the radial expansion by the centrifugal force has taken place and when the magnetic field is homogeneous in the central region of the device. In Fig. 1b. the expansion has moved the inner surface of the plasma in the central region to the radius r_1. If the electrical resistivity of the wall at r_2 is high enough, part of the magnetic flux may diffuse through the wall during the expansion and part of the rotating plasma will be "scraped off" at the wall. The two extreme cases of infinitely-conducting and non-conducting wall are treated here. This is done by introducing a second infinitely-conducting wall at a radial distance R and assuming the resistivity of the chamber wall to be infinite. Then, $R = r_2$ and $R \to \infty$ give the two situations to be studied.

The configuration consists of three regions:

(i) the space inside the inner boundary of the rotating plasma ($r < r_1$),
(ii) the region of the rotating plasma ($r_1 \le r \le r_2$), the only region where volume currents are assumed to flow, and
(iii) the region between the chamber wall and the external wall ($r_2 < r < R$).

Quantities in these three regions will be denoted by superscripts ($'$), ($''$) and ($'''$), respectively.

When no surface currents are assumed to exist at the boundaries between the regions, the magnetic field becomes continuous; this then requires that the following relations are fulfilled:

$$B' = B_0\, b'' - B_0\, a^2/r_1^2, \tag{18}$$

$$B''' = B_0\, b'' - B_0\, a^2/r_2^2. \tag{19}$$

Further, the total magnetic flux inside the infinitely conducting wall at $r = R$ should be conserved, i.e.,

$$R^2\, B_0 = r_1^2\, B' + (r_2^2 - r_1^2)\, B_0\, b'' - 2\, a^2\, B_0 \log (r_2/r_1) + (R^2 - r_2^2)\, B''', \tag{20}$$

where use has been made of Eq. (17). Finally, the magnetic flux inside the boundary $r = r_1$ of the rotating plasma should be conserved since the latter has been assumed to have zero resistivity. This gives

$$\xi_1^2 = r_1^2\, (b'' - a^2/r_1^2). \tag{21}$$

Introduce the notations

$$x = r_1/r_2;\ \ x_0 = a/r_2;\ \ x_1 = \xi_1/r_2. \tag{22}$$

Combination of Eqs. (18), (19), (20) and (21) now yields

$$b' = B'/B_0 = x_1^2/x^2, \tag{23}$$

$$b'' = 1 + x_0^2 + (r_2/R)^2\, x_0^2 \log (x^{-2}), \tag{24}$$

$$b''' = B'''/B_0 = 1 + (r_2/R)^2\, x_0^2 \log (x^{-2}), \tag{25}$$

and

$$x^2 = \frac{x_1^2 + x_0^2}{1 + x_0^2 + (r_2/R)^2 x_0^2 \log (x^{-2})}\,. \tag{26}$$

With the approximations introduced in Section 2.1 the voltage Φ_{12} across the rotating plasma can be found from an integration of Eq. (11) by which

$$\Phi_{12} = \int_{r_1}^{r_2} - v_\varphi\, B dr = r_2\, B_0^2\, \psi_{12}/(2 \mu_0\, \varrho_0)^{\frac{1}{2}} \tag{27}$$

where

$$\psi_{12} = x_0[1 + x_0^2 + (r_2/R)^2 x_0^2 \log (x^{-2})] \log (x^{-2}) - x_0^3\, (x^{-2} - 1) \tag{28}$$

and use has been made of Eqs. (16) and (24).

From Eqs. (15) and (22) the electric charge transmitted per unit length becomes

$$Q = 2\pi r_2 x_0 \left(\frac{2 \varrho_0}{\mu_0}\right)^{\frac{1}{2}}. \tag{29}$$

Eqs. (27) and (29) can be used in a formal deduction of an equivalent capacity per unit length

$$C = \frac{4\pi \varrho_0 x_0}{B_0^2 \psi_{12}} \tag{30}$$

which depends upon the deformation of the magnetic field, upon the transmitted charge Q, and upon the constitution of the chamber wall. Roughly speaking, a considerable deformation of the magnetic field takes place when the parameter x_0 exceeds unity.

In the special situation where the transmitted charge consists of a constant contribution Q_0, together with the charge supplied from an external condenser bank of capacity C_s and initial voltage V_s, the total charge per unit length becomes

$$Q = Q_0 + C_s\,(V_s - \Phi_{12})\,/\,L, \tag{31}$$

where L is the effective axial length of the rotating plasma. Eqs. (29) and (31) yield

$$V_s - \Phi_{12} = (L/C_s)\,[2\pi\, r_2\, x_0\, (2\varrho_0/\mu_0)^{\frac{1}{2}} - Q_0] \tag{32}$$

The voltage characteristics which describe the relation between the voltage V_s of the energy source and the voltage Φ_{12} across the plasma are determined by Eqs. (26), (28), (27) and (32).

Because of diffusion through the chamber wall the magnetic flux Φ_{02} inside $r = r_2$ will be changed by the relative amount

$$\Delta\Phi_{02}/\Phi_{02} = [1 - (r_2/R)^2]\, x_0^2 \log(x^{-2}) \tag{33}$$

as calculated from Eq. (25).

Before a studying a specific example we shall examine two limiting cases:

(i) Small transmitted charge:

When x_0 is much less than unity, x approaches x_1 and there is negligible deformation of the initial field. Then ψ_{12} tends to $x_0 \cdot \log(1/x_1^2)$ and the equivalent capacity is

$$C_0 = \frac{2\pi\varrho_0}{B_0^2 \log(x_1^{-1})} \tag{34}$$

as earlier shown by Anderson *et al.* [3]. The constitution of the wall does not influence the result at small transmitted charges.

(ii) Large transmitted charge:

When x_0 becomes very large, x approaches unity, i.e., the plasma is expanded violently towards the wall at $r = r_2$. A series expansion of Eqs. (26) and (28) then shows that ψ_{12} tends to $(1 - x_1^2)\, x_0$ for an infinitely-conducting wall ($R = r_2$) and to $(1 - x_1^4)/2x_0$ for a non-conducting wall ($R \to \infty$).

When the transmitted charge becomes large the magnetic field is heavily distorted and there is a marked influence by the constitution of the wall. For an infinitely-conducting wall the magnetic field will "lean" against its inner surface and the equivalent capacity tends to

$$C_\infty = \frac{4\pi\varrho_0}{B_0^2(1 - x_1^2)} = \frac{2\,C_0 \log(x_1^{-1})}{(1 - x_1^2)}. \tag{35}$$

As an example $C_\infty/C_0 = 2.29$ for the Ixion where x_1 is about 0.375. The increase in the equivalent capacity is caused by a storage of potential energy in the magnetic field as pointed out by Anderson *et al.* [3], who have deduced Eq. (35) in a different way. All the magnetic and kinetic energy stored in a dissipation-free plasma with an infinitely-conducting boundary can be recovered. However, when the bounding wall has finite conductivity and the magnetic field can diffuse through it, part of the energy will be dissipated when the plasma is "scraped off" the wall and cannot be recovered. In the limit of large transmitted charge it can then be seen from Eqs. (13), (16), (23), (24), (25), (29) and (30) that the kinetic energy within the plasma and the change in magnetic energy both approach the value $\pi\,(1 - x_1^4)\, r_2^2\, B_0^2/2\mu_0$ per unit length however large the passed charge Q becomes.

3. Applications to experiments

3.1. Experimental conditions

The present theory will now be applied to the experiments with the Ixion reported by Boyer *et al.* [4], Baker and Hammel [8, 9], and Baker and Ribe [10]. According to Baker and Hammel [11] a cylindrical plasma beam with an effective diameter of 0.09 m acts as centre electrode. The diameter of the cylindrical wall of the chamber is 0.24 m and the length 0.9 m. Consequently, we put $\xi_1 = 0.045$ m, $r_2 = 0.12$ m and $x_1 = 0.375$. The magnetic field at the surface of the end insulators is 2.2 times stronger than that at the median plane.

Boyer *et al.* report an experiment where deuterium is injected corresponding to a pressure of $1.5 \times \times 10^{-3}$ mmHg and that impurities of H, C^{II}, Fe^{I}, Fe^{II}, Si^{III}, Si^{II} and O^{II} are released from the walls at an early stage of the discharge and take part in the rotation. The amount of the impurities corresponds to about 10^{-3} mmHg in the experiment described by Boyer *et al.*, i.e., about 40 % of the total number of particles. Impurities contribute to the density also in later experiments with the Ixion [11].

The present theory for the balance of forces will now be applied to the median plane of the Ixion where the magnetic field is nearly homogeneous. In order to determine the actual density within this region of the rotating plasma we have to estimate the effective axial length L. The axial contraction of the plasma towards the median plane produced by the centrifugal force then has to be taken into account. As shown by the author in an earlier paper [12] this contraction can become considerable when the mirror ratio of the device is large.

An estimate of the actual density in the median plane can be made by applying the present theory to the measurement by Baker and Ribe [10] on the diamagnetic effects of the Ixion. With a capacity $C_s = 140\,\mu F$, a voltage $V_s = 7.5$ kV, a drop to $\Phi_{12} \approx \frac{1}{3} V_s$ when B_0 is about 0.7 V sec/m² the diamagnetic effect gives a field reduction $b' = 0.6$ at the centre of the device. Then, $x^2 = 0.234$ according to Eq. (23). This gives $x_0^2 = 0.219$ for an infinitely-conducting wall and $x_0^2 = 0.122$ for a non-conducting wall and, according to Eq. (32), $L^2\,\varrho_0 \approx 1.4 \times 10^{-6}$ and 2.5×10^{-6} kg/m³, respectively. In the experiment by Baker and Ribe there is a considerable deformation of the magnetic field within an axial length of about 0.2 m. Assuming the main part of the plasma to be situated within this region by putting $L = 0.2$ m, we obtain $\varrho_0 \simeq 3.5 \times$ $\times 10^{-5}$ and 6.2×10^{-5} kg/m³, respectively.

With the data given by Baker and Hammel [9, 11] for the experiment on voltage limiting in the Ixion, the deuterium has a density of 5×10^{-6} kg/m³ before axial contraction. Then $\varrho_0 \simeq 5 \times 10^{-5}$ kg/m³ if $L = 0.2$ m and 20% impurities of a mean mass number of 30 are assumed to be present. The equations determining the voltage characteristics are not very sensitive to changes in the density. In the following discussion we put $\varrho_0 = 5 \times 10^{-5}$ kg/m³ in the median plane and $L = 0.2$ m.

3.2. Voltage characteristics of the Ixion

In the experiments on voltage limiting in the Ixion [9, 11] a first bank of 120 microfarads capacity and 7 kilovolts initial voltage was connected to the device to produce a rotating plasma. After the rotation had been established the first bank was disconnected and a second bank of capacity $C_s = 149\,\mu F$ was immediately switched in its place. The voltage V_s of this second bank could be varied up to 20 kilovolts and an external series resistance of 1.33 ohms was used to limit the current through the plasma. In the experiments a voltage increase was observed, the peak value Φ_{12} of which is shown as a function of the bank voltage V_s by the dashed lines in Figs. 2 and 3. As pointed out by Baker and Hammel [9] the peak voltage across the plasma shows a tendency to saturate when the second bank voltage is increased up to 20 kV. After the peak the voltage decreases and passes the value before the connection of the second bank after some 50 to 100 μ sec.

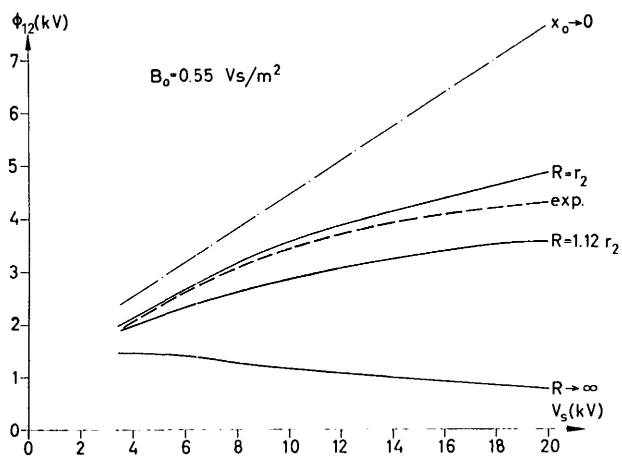

Fig. 2 Voltage Φ_{12} across the rotating plasma in the Ixion as a function of the second capacitor bank voltage V_s. The value of B_0 is 0.55 Vs/m². Dashed curves give experimental results for the peak values of Φ_{12} according to Baker and Hammel [9]. Full curves give calculated voltage characteristics at full charge transfer for an assumed plasma density $\varrho_0 = 5 \times 10^{-5}$ kg/m³ and an effective axial length $L = 0.2$ m. An infinitely conducting chamber wall corresponds to $R = r_2$ and a non-conducting wall to $R \to \infty$. The radial diffusion of the magnetic flux across the wall at $r = r_2$ can be simulated by a proper choice of R. The intermediate full curves demonstrate the case $R = 1.12\, r_2$. Dot-and-dash lines give the characteristics which would have been obtained for a field of infinite rigidity.

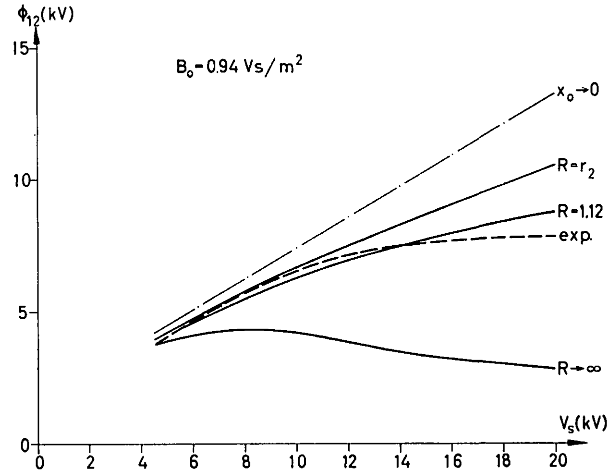

Fig. 3 Voltage Φ_{12} across the rotating plasma in the Ixion as a function of the second capacitor bank voltage V_s. Same as Fig. 2, but with $B_0 = 0.94$ Vs/m².

By means of Eqs. (27), (28), (31) and (32) the voltage Φ_{12} across the plasma has been calculated as a function of the second bank voltage V_s for the magnetic field strengths 0.55 and 0.94 V sec/m². The calculation applies to the situation where the charge has been shared between the bank and the plasma. The results which are shown by Figs. 2 and 3 have been determined from the data of this Section 3.2, where the first bank determines the initial charge Q_0. The full curves demonstrate the cases of infinitely-conducting chamber wall ($R = r_2$), non-conducting wall ($R \to \infty$) and the situation with $R^2/r_2^2 = 1.25$ which allows part of the magnetic flux to leak through the wall at $r = r_2$. Dot and dash lines give the characteristics which would have been obtained without any deformation of the magnetic field.

4. Discussion

The equivalent circuit of a rotating plasma device consists of a capacity, a parallel resistance and an internal series impedance [5]. The latter is determined by the resistivity of the plasma, the Hall current, the presence of neutral gas and by the state of motion. A determination of the magnitude of this impedance is not within the scope of the present paper. We shall only discuss the situation where inductances as well as charge leakage across the parallel resistance can be neglected. There are two possibilities:

(i) The internal series impedance is negligible. The largest voltage which can occur across the plasma is then reached when the charge has been shared between the plasma and the condenser bank.

(ii) The internal series impedance is comparable to or exceeds the external series impedance. A jump of the voltage across the plasma then occurs when the second bank is connected, the jump being determined by the ratio between the internal and external impedances. If this ratio becomes very large the voltage jump may even exceeds the equilibrium value which is approached as soon as the charge has been shared between the second bank and the plasma.

A comparison between the voltage characteristics of Fig. 2 obtained from the Ixion experiments and the calculated curves shows that a possible explanation of the saturation can be provided by the present theory. A very large ratio between the internal and external series impedances should give a voltage jump nearly equal to the second bank voltage; this seems to be ruled out by the experiments. If the same ratio is not too large, however, a saturation of the voltage will be produced by the increase of the equivalent capacity through deformation of the magnetic field, as soon as the charge has been transmitted to the Ixion. In Figs. 2 and 3 the theoretical curves for an infinitely-conducting wall show a strong reduction of the plasma voltage as compared to a situation (the dot and dash lines) corresponding to a magnetic field of infinite rigidity. Observe that the reduction is larger in Fig. 2 than in Fig. 3; the magnetic field is weaker in the former figure than in the latter. For a non-conducting wall the same figures show that the plasma voltage would even drop far below the observed experimental values.

A rough estimate of the field diffusion through the chamber wall can be made by considering the time constant for a circulating azimuthal wall current. With a thickness d_2 and conductivity σ_2 of the wall the time constant becomes about

$$\tau_2 = \tfrac{1}{3}\mu_0\,\sigma_2\,r_2\,d_2. \tag{36}$$

In the experiments on voltage limiting a stainless steel wall of $d_2 = 1.78 \times 10^{-3}$ m thickness has been used [11]. With $\sigma_2 = 1.6 \times 10^6\ \Omega^{-1}\,\mathrm{m}^{-1}$ this gives $\tau_2 \simeq 210\ \mu$ sec.

For the bank voltage $V_S = 20$ kV and the conditions applying to Figs. 2 and 3. Eq. (33) would give about 70% reduction of the magnetic flux inside the device if the chamber wall is non-conducting. In the actual case, connection of the second bank gives an increase in the Ixion voltage which lasts for about $30\ \mu$ sec. The current induced in the wall by the expanding flux will decay by about 15% when the same time has elapsed. The effect of a flux reduction of 15% by diffusion through the wall can then be simulated by putting $R = 1.12\ r_2$ which gives the intermediate full curves of Figs. 2 and 3. From these curves is seen that even a slight diffusion of the magnetic field through the chamber wall in an early stage of the discharge reduces the calculated plasma voltage to values nearly equal to the observed ones.

As time is proceeding the magnetic field continues to diffuse through the wall and the plasma approaches a state corresponding to a non-conducting wall after a time of the order of τ_2. The voltage across the device will then decrease appreciably and part of the plasma will be "scraped off" at the wall. This gives one possible explanation of the observed decrease of the voltage [9], about 100 μ sec. after the connection of the second bank. In particular, it would be interesting to see if the voltage characteristics can be modified by the presence of a chamber wall made of a highly conducting material such as copper.

Finally, if there is an increase in the amount of impurities when the bank voltage is increased, then the density of the plasma may become an increasing function of the voltage. This would also contribute to the effects which produce a saturation in the voltage characteristics.

Acknowledgements

The author wants to express his sincere thanks to Drs. D. A. Baker and J. E. Hammel of Los Alamos Scientific Laboratory for supplying a number of data on the Ixion device.

References

[1] Early, H. C., and Dow, W. G., *Phys. Rev.* **79** (1950) 186.

[2] Anderson, O. A., Baker, W. R., Bratenahl, A., Furth, H. P., Ise, J., Kunkel, W. B., and Stone, J. M., *Proc. Second U. N. Int. Conf. on the Peaceful Uses of Atomic Energy,* **32** (1958) 155.

[3] Anderson, O. A., Baker, W. R., Bratenahl, A., Furth, H. P., and Kunkel, W. B., *J. Appl. Phys.* **30** (1959) 188.

[4] Boyer, K., Hammel, J. E., Longmire, C. L., Nagle, D., Ribe, F. L., and Riesenfeld, W. B., *Proc. Second U. N. Int. Conf. on the Peaceful Uses of Atomic Energy,* **31** (1958) 319.

[5] Lehnert, B., *Progress in Nuclear Energy*: Series XI: Vol. 2, "Plasma Physics and Thermonuclear Research" (J. L. Tuck, Ed.: Pergamon Press 1961).

[6] Alfvén, H., Cosmical Electrodynamics, (Clarendon Press, Oxford, 1950) p. 50.

[7] Walén, C., *Arkiv Mat. Astron. Fysik* **33 A,** No. 18 (1946).

[8] Baker, D. A., and Hammel, J. E., *Bull. Amer. Phys. Soc.*, Ser. II, **5** (1959) 341.

[9] Baker, D. A., and Hammel, J. E., *Bull. Amer. Phys. Soc.*, Ser. II, **5** (1959) 314.

[10] Baker, D. A., and Ribe, F. L., *Bull. Amer. Phys. Soc.*, Ser. II, **4** (1959) 248.

[11] Baker, D. A., and Hammel, J. E., Private communication (1960).

[12] Lehnert, B., *Arkiv f. Fysik,* **18** (1960) 251; *Rev. Mod. Phys.* **32** (1960) 1012.

(Manuscript received on 12 November 1960)